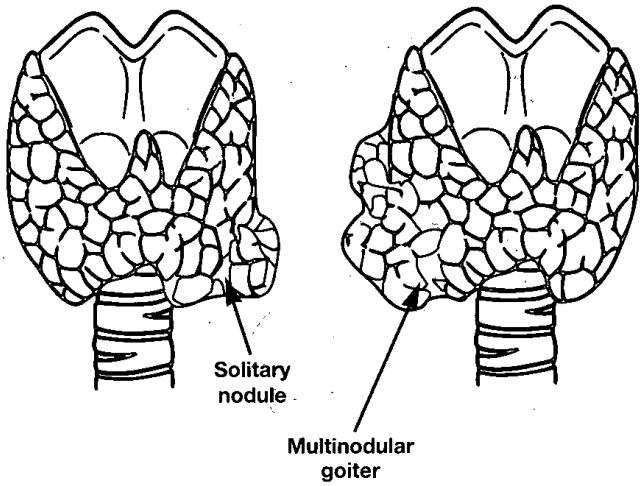
Thyroid Nodules

Thyroid nodules are lumps in your thyroid gland. You may have one nodule, called a **solitary nodule**, or you can have several, which is a condition called **multinodular goiter**. Thyroid nodules are often harmless, and they tend to run in families.



Thyroid nodules are rarely cancerous. Sometimes, nodules produce too much thyroid hormone and cause hyperthyroidism. If you have thyroid nodules, there are a variety of treatment options available, although sometimes close observation without any treatment is preferred.

If you would like more information on thyroid nodules, pick up a copy of *Understanding Thyroid Nodules.*

Making a Diagnosis

If you have a goiter, it may look like your neck has gotten wider, or you may have more swelling on one side. Often, there are no symptoms of a goiter. However, some possible signs and symptoms of a goiter are:

- visible enlargement
- pain in your throat
- twinges of pain in your throat with swelling
- hoarseness
- difficulty or discomfort in swallowing

If you have hyperthyroidism, you may have a goiter, feel jittery, experience weight loss, be less tolerant to heat, and sweat more than usual. That is because the extra thyroid hormone in your blood is **speeding up** your entire body. Left untreated, hyperthyroidism can cause serious damage to your health, so it is very important that the disease be identified early and treated properly.

For more information on hyperthyroidism, pick up a copy of *Hyperthyroidism and Your Health.*

Hyperthyroidism

Hyperthyroidism occurs when there is too much thyroid hormone circulating in your body. It can be caused by several things, including

- ***Graves' Disease***

Graves' disease is a condition in which the **immune system** causes too much thyroid hormone to be made and released into the blood

- ***Thyroid Nodules***

Thyroid nodules are **lumps** in the thyroid gland that may sometimes overproduce thyroid hormone

- ***Overreplacement of Thyroid Hormone***

A patient receiving **thyroid hormone replacement therapy** for hypothyroidism and given too high a dose (or too much thyroid hormone) can become hyperthyroid

- ***Inflammation of the Thyroid***

Inflammation of the thyroid can cause damage to thyroid tissue

If your goiter is caused by an underlying thyroid disorder (ie, hypothyroidism or hyperthyroidism), you may also experience the signs and symptoms of that disorder.

If you or your doctor suspect a thyroid disorder or goiter, your doctor will probably investigate the following things.

Your Personal and Family Health History

Thyroid disorders can often be **hereditary**. If you have a family member, like your mother or father, an aunt or uncle, or your brother or sister, who has a thyroid disorder, your chance of developing one may be higher.

Your Likelihood of Developing a Disorder

Some individuals are more likely than others to develop thyroid disorders. Those more likely than others to become hypothyroid include:

- ✔ women, especially those over 40
- ✔ postpartum women (within 6 months after delivery)
- ✔ people with high cholesterol
- ✔ people experiencing depression
- ✔ people who have had thyroid disease before
- ✔ people with a family history of autoimmune thyroid disease
- ✔ people who have been treated for *hyper*thyroidism
- ✔ people who have had x-ray therapy of the head or neck for cancers or Hodgkin's disease
- ✔ people who have taken or who are taking lithium or amiodarone
- ✔ people with nonthyroid autoimmune conditions, for example:

  —Addison's disease
  —Type I diabetes
  —Pernicious anemia

You may notice symptoms such as tiredness, intolerance to cold, or depression if you have hypothyroidism. What is happening is that the low levels of thyroid hormone are causing an overall slowing down of your body's functions. Over time, this can cause serious health problems, which is why it is essential that hypothyroidism be identified and treated quickly.

For more information on hypothyroidism, pick up a copy of *Hypothyroidism and Your Health* or *What Is Hashimoto's Disease?*

A goiter can also be caused by

- **Hypothyroidism** (underactivity of the thyroid)
- **Hyperthyroidism** (overactivity of the thyroid)
- **Thyroid nodules** (lumps in the thyroid)
- **Inflammation of the thyroid gland** (possibly due to infection)

Hypothyroidism

If you have too little thyroid hormone in your bloodstream—even after your brain has signaled the thyroid to "turn up the heat" and produce more—you may have a condition called **hypothyroidism**.

Hypothyroidism is commonly caused by **Hashimoto's disease**, a condition in which the immune system interferes with normal thyroid gland function. Other causes of hypothyroidism are thyroid surgery and treatment for hyperthyroidism.

—Rheumatoid arthritis
—Systemic lupus erythematosus (SLE or lupus)

Those more likely than others to become hyperthyroid...

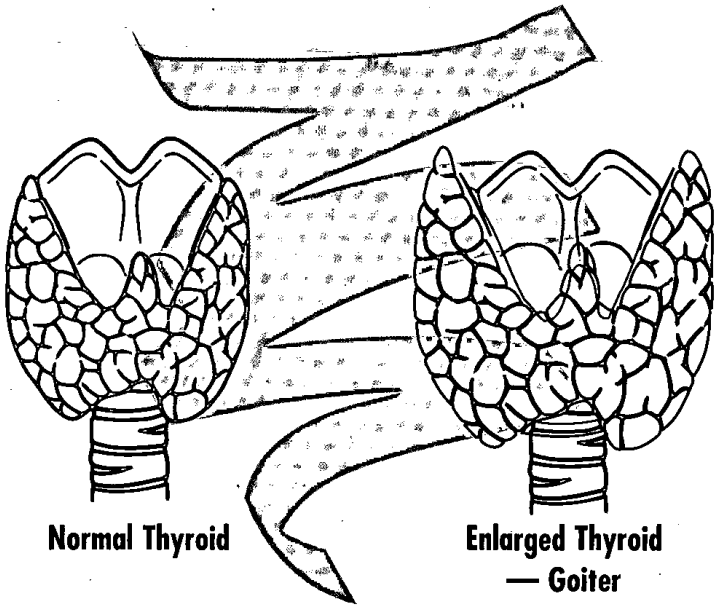
- ☑ are women, 30 years or older
- ☑ are pregnant or have given birth in the last 6 months*
- ☑ have other autoimmune conditions, such as:
  - —Addison's disease
  - —Type I diabetes
  - —Pernicious anemia
  - —Rheumatoid arthritis
  - —Systemic lupus erythematosus (SLE or lupus)

***Note:** If you've had a baby in the past 6 months, you may be at a higher risk for developing hyperthyroidism due to autoimmune thyroid disease.

Your Physical Condition

- Your doctor will probably give you a physical exam to look for the signs and symptoms of thyroid disorders

- In particular, your doctor will carefully examine your thyroid gland. Your doctor will press on, or **palpate**, the area of your neck where your thyroid is located to determine if a goiter or thyroid nodules are present

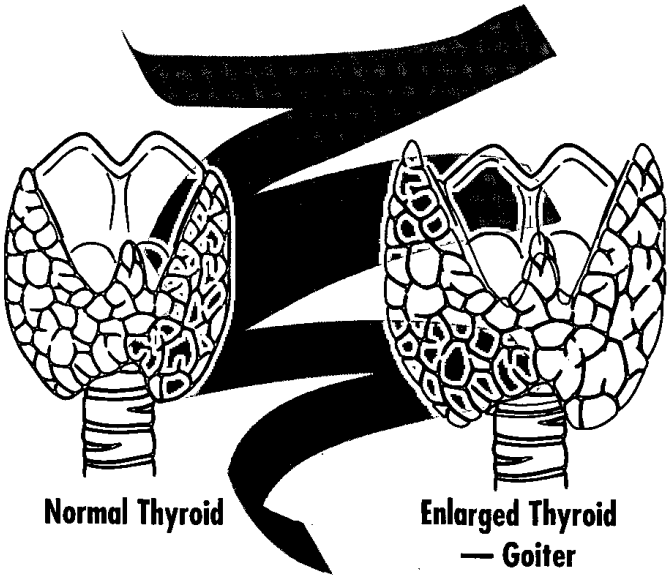


What Can Cause A Goiter?

Basically, many problems with your thyroid gland can cause a goiter. Worldwide, the most common cause of a goiter is a **lack of iodine** in the diet. Iodine is an essential part of the process of making thyroid hormone. Without enough iodine, the thyroid gland cannot produce enough thyroid hormone to meet the body's needs. It then becomes enlarged when TSH tries to stimulate thyroid hormone production. In the United States, however, iodine has been added to table salt, which is a basic part of the diet of most Americans. For this reason, the US has nearly eliminated thyroid disorders caused by lack of iodine.

Goiter— A Closer Look

The term **goiter** simply means an enlarged thyroid gland. Your thyroid gland may be enlarged because it's not working properly, or it can be enlarged temporarily for a variety of reasons, and may shrink back to normal by itself. Although a goiter is usually a reliable sign of a thyroid disorder, it doesn't always need to be treated.



Your Laboratory Results

Your doctor will likely order blood tests to find out how your thyroid gland is functioning. The best single test is the **TSH test**. Your doctor may also measure your **thyroid hormone levels** as well as your TSH level. These tests can confirm the presence of hypothyroidism or hyperthyroidism.

There are additional tests that your doctor may order for you. Among them is a **thyroid antibody test**, which can help determine if you have Hashimoto's disease.

There are also a number of procedures followed to evaluate nodules, including thyroid scans, ultrasound, and fine-needle biopsy.

For more complete information regarding these procedures, pick up a copy of *Understanding Thyroid Nodules.*

Treatment Options for Goiter

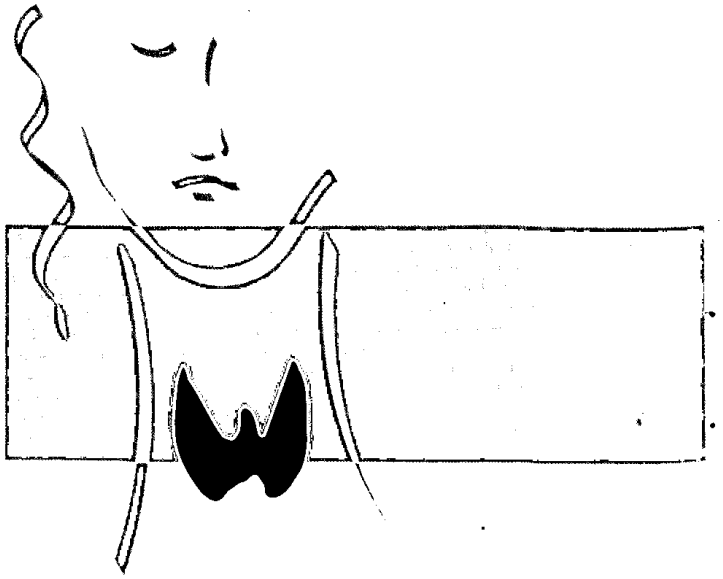
If you have a goiter that is caused by hypothyroidism or hyperthyroidism, your doctor will treat the underlying disorder, thereby reducing the goiter.

If you have a noncancerous goiter that is not caused by hypo- or hyperthyroidism, your doctor will determine a course of treatment based on your specific needs and overall health.

If your doctor thinks treatment is appropriate, you may be given **thyroid hormone therapy**, which may help shrink the goiter. However, many people do not benefit from this therapy, especially people who have had untreated goiters for many years.

Your pituitary then senses that there is just the right amount of thyroid hormone in your body. With your thyroid hormone levels now restored to normal, your pituitary slows its production of TSH back down to normal.

For instance, just as the thermometer in the thermostat senses the temperature of a room, your **pituitary** constantly senses the amount of thyroid hormone in your blood. If there is not enough thyroid hormone, your pituitary senses the need to "turn on the heat." It does this by releasing more **Thyroid-Stimulating Hormone (TSH)**, which signals your thyroid to make more thyroid hormone. Your thyroid gland then makes and releases the hormone directly into your bloodstream and restores the amount of thyroid hormone in your blood to normal.



Another treatment option is surgery, which is occasionally done to remove the goiter if it is putting pressure on the throat, or if it is causing cosmetic concerns.

Whether the decision is to treat or not to treat, your doctor will probably monitor your thyroid gland periodically to ensure that if you develop additional problems, they can be treated promptly.

Additional information about nodules and thyroid cancer is available in the booklets *Understanding Thyroid Nodules* and *Understanding Thyroid Cancer.*

A Final Note

Your thyroid gland plays a key role both in your general health and in your daily activities. Being aware of the different types of thyroid conditions and their possible signs and symptoms is an excellent way to prevent the long-term, damaging effects of leaving a thyroid disorder untreated. If you think you have a thyroid disorder, speak with your doctor about your concerns.

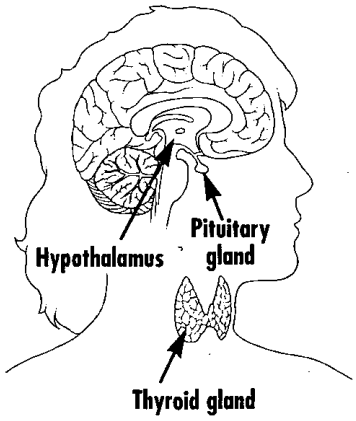
If you are diagnosed with a thyroid-related problem, be assured that there are excellent treatments available to you. Find out as much as you can about your condition, and talk to your doctor about the need and options for treatment. By following your doctor's instructions, you can lead a normal, healthy life.

The amount of thyroid hormone made by your thyroid gland is adjusted by another gland called the **pituitary**, as well as a part of your brain called the **hypothalamus**.

- The pituitary gland makes hormones that control many other glands in your body

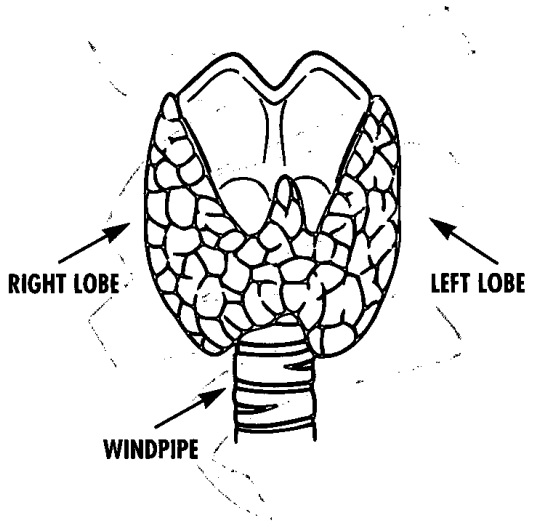
- The hypothalamus helps the pituitary control many of your glands, as well as helping to control other bodily functions, such as thirst, hunger, sleep, and body temperature

The thyroid gland, pituitary gland, and hypothalamus all work together to control the amount of thyroid hormone in your body. With the pituitary controlling most of the action, these structures work in a way that is similar to the way a thermostat controls the temperature in a room.



Your Thyroid— The Basics

Your thyroid is a gland located at the base of your neck, just below your Adam's apple. It is shaped like a butterfly—each "wing," or **lobe**, of your thyroid lies on either side of your windpipe.



The purpose of your thyroid gland is to make, store, and release **thyroid hormone** into your blood. Thyroid hormone affects essentially every cell in your body, and helps control your body's functions. If you have too little thyroid hormone in your blood, your body slows down. This is a condition called **hypothyroidism**. If you have too much thyroid hormone in your blood, your body speeds up. This is a condition called **hyperthyroidism**.

For more information on the thyroid gland, please contact:

The Thyroid Foundation of America
Ruth Sleeper Hall, RSL 350
40 Parkman Street
Boston, MA 02114-2698
(send self-addressed, stamped envelope)

The Thyroid Society for Education & Research
7515 S. Main Street, Suite 545
Houston, TX 77030
1-800-THYROID

National Graves' Disease Foundation
2 Tsitsi Court
Brevard, NC 28712

Or visit our web site at:
www.glandcentral.com

Notes:

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Goiter—
What You Need to Know

A goiter is an enlarged thyroid gland. The thyroid may be enlarged because it's not working properly, or it can be enlarged for other reasons. Although a goiter is usually a reliable sign of a thyroid disorder, it doesn't always need to be treated.

The following pages contain basic information about the thyroid gland, its role in the body, and thyroid disorders. In addition, this booklet will discuss the causes and treatment of a goiter.

Remember, no single booklet can answer all of your questions or replace speaking with doctors, nurses, and other members of the healthcare team. If you have any questions or concerns about the thyroid or thyroid disorders, please check with your doctor.

Notes:

This booklet is part of a continuing patient education series on the thyroid gland and thyroid disease. Additional booklets available in this series are:

Getting to Know Your Medication
How the Thyroid Gland Can Affect Your Weight
Hyperthyroidism and Your Health
Hypothyroidism and Your Health
The Postpartum Period and Your Thyroid
The Thyroid Gland and Your Health
Thyroid Disease and Aging
Thyroid Disease and Diabetes
Thyroid Disease in Women
Understanding Graves' Disease
Understanding Thyroid Cancer
Understanding Thyroid Nodules
Understanding TSH Testing
Understanding Your Child's Thyroid
What Is Hashimoto's Disease?

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Knoll Pharmaceutical Company
3000 Continental Drive - North
Mount Olive, NJ 07828-1234

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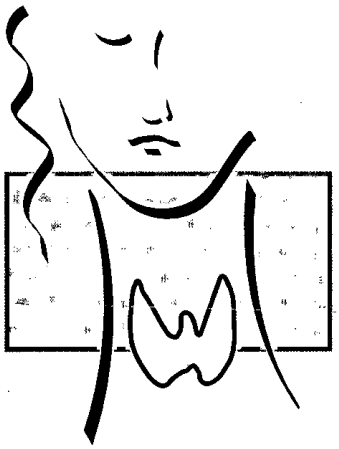
Understanding Goiter